

Numerical Weather Prediction: Advancements and AI Integration

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Abstract—Numerical Weather Prediction (NWP) has undergone significant advancements since its inception, transforming from rudimentary empirical methods to sophisticated computational models that simulate the complex dynamics of the Earth’s atmosphere. This paper provides a comprehensive overview of modern NWP systems, detailing their underlying principles, the intricate process of data assimilation, and the computational challenges inherent in producing accurate and timely forecasts. We explore the foundational mathematical equations that govern atmospheric processes and the numerical techniques employed for their discretization. Furthermore, we delve into the burgeoning field of Artificial Intelligence (AI) and Machine Learning (ML) in NWP, examining how these innovative approaches are being leveraged to enhance forecast accuracy, computational efficiency, and the parameterization of sub-grid scale phenomena. This includes a discussion on purely data-driven weather models, hybrid physics-informed neural networks, and their comparative performance against traditional physics-based models. The paper also highlights the current limitations and future directions, emphasizing the potential for synergistic integration of physical models with AI to unlock new frontiers in weather forecasting, from short-term nowcasting to long-range seasonal predictions.

Index Terms—Numerical Weather Prediction, Machine Learning, Deep Learning, Atmospheric Modeling, Data Assimilation, Ensemble Forecasting, Physics-Informed Machine Learning

I Introduction

Numerical Weather Prediction (NWP) stands as a cornerstone of modern meteorology, providing the essential forecasts that impact nearly every aspect of human life, from daily planning and agricultural decisions to disaster preparedness and aviation safety [1]. The ability to predict future atmospheric states relies on the integration of physical principles, sophisticated mathematical models, vast amounts of observational data, and immense computational power. Since the pioneering work of Lewis Fry Richardson in the early 20th century, NWP has evolved dramatically, moving from manual computations to highly complex global and regional models run on supercomputers [2].

The fundamental premise of NWP involves solving a set of partial differential equations (PDEs) that describe the conservation of momentum, energy, and mass in the atmosphere, coupled with the ideal gas law and equations for atmospheric moisture. These equations are

highly nonlinear and computationally intensive, requiring numerical approximation techniques for their solution [3]. The process begins with the assimilation of observational data from a diverse range of sources, including satellites, ground stations, radar, and weather balloons, to define the initial state of the atmosphere as accurately as possible. This initial state is then used by the NWP model to project atmospheric conditions forward in time.

Despite remarkable progress, NWP still faces significant challenges. The inherent chaotic nature of the atmosphere, as famously described by Edward Lorenz, imposes fundamental limits on predictability [4]. Furthermore, the computational cost of running high-resolution models, the accurate representation of sub-grid scale physical processes (e.g., clouds, turbulence), and the uncertainties in initial conditions continue to be active areas of research. These challenges have spurred interest in alternative and complementary approaches, particularly those offered by Artificial Intelligence (AI) and Machine Learning (ML) [5], [6].

Recent breakthroughs in deep learning have demonstrated the capability of AI models to learn complex patterns from massive meteorological datasets, offering potential avenues to enhance traditional NWP systems. These AI applications range from improving specific components of NWP, such as parameterizations and data assimilation, to developing entirely data-driven end-to-end forecasting models that can provide forecasts with unprecedented speed [7]–[9]. The synergy between physics-based modeling and data-driven AI approaches is becoming increasingly crucial, promising to overcome some of the long-standing hurdles in weather forecasting [10].

This paper is structured as follows: Section II reviews the historical context and recent advancements in both traditional NWP and AI-driven weather prediction. Section III elaborates on the core components of NWP, including the governing equations, numerical methods, and data assimilation, and details how AI/ML techniques are being integrated. Section IV presents a discussion of the performance and comparative advantages of these different approaches, highlighting their strengths and limitations. Finally, Section V concludes the paper with a summary of key findings and outlines future research

directions in this dynamic field.

II Related Work

The field of Numerical Weather Prediction has a rich history, evolving over decades from theoretical concepts to operational systems. Early work laid the mathematical foundations, while advancements in computing power enabled increasingly complex simulations. More recently, Artificial Intelligence and Machine Learning have emerged as powerful tools, creating a new paradigm for weather forecasting.

II-A Traditional NWP Models

Traditional NWP models are founded on a set of fundamental atmospheric equations that describe the conservation of momentum, energy, and mass. Pioneering efforts by Richardson in the 1920s demonstrated the theoretical possibility of NWP, though practical implementation awaited the advent of electronic computers [1]. Post-World War II, the development of numerical methods and the first digital computers led to the establishment of the first operational NWP centers. Key milestones include the development of quasi-geostrophic models and later, primitive equation models, which more accurately represent atmospheric dynamics [2]. These models divide the atmosphere into a three-dimensional grid, numerically solving prognostic equations for atmospheric variables at each grid point and then advancing these solutions over discrete time steps.

Modern operational NWP centers, such as the European Centre for Medium-Range Weather Forecasts (ECMWF) and the National Centers for Environmental Prediction (NCEP) in the USA, employ highly sophisticated global and regional models. Examples include the ECMWF’s Integrated Forecasting System (IFS), the Global Forecast System (GFS) from NCEP, and regional models like the Weather Research and Forecasting (WRF) model [1]. These models feature advanced physics packages for processes like cloud microphysics, radiation, boundary layer turbulence, and convection, which occur at scales smaller than the model grid resolution and thus require parameterization [11]. The ongoing push for higher spatial and temporal resolution in these models increases their computational demand exponentially, necessitating supercomputing resources.

II-B Data Assimilation Techniques

A crucial component of NWP is data assimilation, the process of optimally combining observations from diverse sources with short-range forecasts to produce the best possible estimate of the current atmospheric state, known as the analysis [3]. The quality of this initial state is paramount, as even small errors can grow rapidly due to the chaotic nature of the atmosphere. Early methods, such as optimal interpolation, provided a statistical framework for combining observations and forecasts. However, they

were limited in their ability to account for flow-dependent error correlations.

More advanced techniques like three-dimensional variational (3D-Var) and four-dimensional variational (4D-Var) data assimilation became standard for global models due to their ability to account for flow-dependent error covariances and incorporate observations over a time window [12]. These methods minimize a cost function that measures the misfit between the model state, the background forecast, and the observations. Ensemble Kalman Filter (EnKF) methods offer an alternative by explicitly representing forecast uncertainties through an ensemble of model states, which is particularly effective for highly nonlinear systems and flow-dependent error statistics [13]. The computational expense of data assimilation, especially 4D-Var and EnKF, is substantial, often consuming a significant portion of the total NWP system’s computational budget.

II-C AI and Machine Learning in NWP

The rise of AI and ML, particularly deep learning, has opened new avenues for weather prediction, driven by advancements in algorithms, computational hardware, and the availability of large datasets. Initial applications focused on post-processing NWP outputs, such as bias correction, downscaling, and improving statistical relationships to enhance local forecasts [14]. However, recent advancements have expanded the scope of AI in NWP considerably, moving towards more fundamental integrations.

II-C1 Machine Learning for Parameterization

One significant area of research involves using ML to replace or augment traditional physics parameterizations for sub-grid scale processes. Deep neural networks can learn complex relationships from high-resolution simulations or observations to develop more accurate and computationally efficient parameterizations for convection, clouds, radiation, and boundary layer processes [15], [16]. This approach aims to address a major bottleneck in NWP, where traditional parameterizations are often computationally expensive, rely on simplifying assumptions, and are a primary source of model error. ML-based parameterizations can potentially capture more intricate non-linear relationships than traditional schemes, leading to improved representation of atmospheric phenomena.

II-C2 Data-Driven End-to-End Forecasting Models

More ambitious approaches involve developing entirely data-driven models that learn to predict future atmospheric states directly from past observations, bypassing the explicit solution of physical equations. Models like FourCastNet [?], Pangu-Weather [8], and GraphCast [9] leverage massive reanalysis datasets (e.g., ERA5) and advanced deep learning architectures (e.g., Fourier Neural Operators, U-Nets, Graph Neural Networks, Transformers) to produce global medium-range forecasts. These models have demonstrated competitive or even superior

accuracy to state-of-the-art physics-based models for certain variables and lead times, often with significantly reduced computational cost during inference [17]. Their speed enables rapid iteration and the generation of large ensembles, which is crucial for uncertainty quantification [18].

II-C3 Hybrid Physics-Informed Approaches

Recognizing the strengths of both physics-based and data-driven models, hybrid approaches seek to combine their best attributes. Physics-informed neural networks (PINNs) integrate physical laws directly into the learning process by adding terms to the loss function that penalize deviations from known governing equations [19], [20]. Other hybrid models use ML to learn model error, accelerate specific physical components within a traditional NWP framework, or couple ML models with NWP models to improve stability and physical consistency while retaining computational efficiency [21]. This includes utilizing ML to predict model biases or to refine the outputs of physical parameterizations.

II-C4 AI for Nowcasting and Extreme Events

AI has also shown great promise in nowcasting (short-term forecasting, typically 0-6 hours) by identifying rapidly evolving local weather phenomena, such as heavy precipitation and thunderstorms, from high-resolution radar and satellite data [22]. Convolutional LSTMs and other spatio-temporal networks are particularly effective here. For extreme weather events, ML models are being developed to identify precursors, improve intensity and track prediction for tropical cyclones, and forecast heatwaves or droughts [23]. These applications often leverage spatio-temporal deep learning architectures to capture the dynamic evolution of such events. Recent advances indicate potential for AI in predicting high-impact weather phenomena [24].

The integration of AI into NWP represents a paradigm shift, promising to overcome long-standing limitations through enhanced efficiency, improved accuracy, and the ability to model complex, multiscale interactions more effectively. The ongoing research focuses on bridging the gap between the interpretability and physical consistency of traditional models and the speed and pattern recognition capabilities of AI [25], [26].

III Methodology

The methodology of Numerical Weather Prediction encompasses a series of complex steps, from formulating the governing equations to assimilating observations and executing prognostic models. The integration of AI/ML techniques introduces new facets to this established framework, offering both augmentations and alternative pathways for forecasting.

III-A Foundational Principles of Traditional NWP

Traditional NWP models are built upon a set of primitive equations derived from fundamental laws of physics applied to the atmosphere. These equations describe the conservation of momentum, energy, mass, and water vapor for a fluid on a rotating sphere.

III-A1 Governing Equations

The atmosphere's state is typically described by prognostic variables such as horizontal wind velocity components (u, v), vertical velocity (w or dp/dt), temperature (T), surface pressure (P_s), and specific humidity (q). The full set of primitive equations, often simplified for atmospheric modeling (e.g., hydrostatic approximation), includes:

- 1) Horizontal Momentum Equations: Describe the change in horizontal velocity due to pressure gradient forces, Coriolis force, and frictional forces.

$$\begin{aligned}\frac{Du}{Dt} - fv &= -\frac{1}{\rho} \frac{\partial p}{\partial x} + F_x \\ \frac{Dv}{Dt} + fu &= -\frac{1}{\rho} \frac{\partial p}{\partial y} + F_y\end{aligned}$$

Here, $D/Dt = \frac{\partial}{\partial t} + u \frac{\partial}{\partial x} + v \frac{\partial}{\partial y} + w \frac{\partial}{\partial z}$ is the material derivative, $f = 2\Omega \sin \phi$ is the Coriolis parameter (Ω is Earth's angular velocity, ϕ is latitude), ρ is air density, p is pressure, and F_x, F_y represent frictional forces (e.g., turbulence).

- 2) Vertical Momentum Equation (Non-hydrostatic): Accounts for vertical accelerations. In hydrostatic models, this simplifies to hydrostatic balance ($-\frac{1}{\rho} \frac{\partial p}{\partial z} - g = 0$).

$$\frac{Dw}{Dt} = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g + F_z$$

where g is the acceleration due to gravity, and F_z is vertical friction.

- 3) Thermodynamic Equation (First Law of Thermodynamics): Describes temperature change due to adiabatic processes, diabatic heating (radiation, latent heat release), and turbulent mixing.

$$\frac{DT}{Dt} - \frac{1}{\rho c_p} \frac{Dp}{Dt} = \frac{Q}{c_p}$$

where c_p is the specific heat at constant pressure and Q is the diabatic heating rate per unit mass.

- 4) Continuity Equation: Describes the conservation of mass for a compressible fluid.

$$\frac{D\rho}{Dt} + \rho(\nabla \cdot \mathbf{V}) = 0$$

where $\mathbf{V} = (u, v, w)$ is the three-dimensional velocity vector.

- 5) Equation of State (Ideal Gas Law): Relates pressure, density, and temperature.

$$p = \rho R_d T_v$$

where R_d is the gas constant for dry air and T_v is the virtual temperature, which accounts for moisture effects on density.

- 6) Moisture Equation: Describes the conservation of water vapor and its phase changes (condensation, evaporation, sublimation).

$$\frac{Dq}{Dt} = S_q$$

where q is specific humidity and S_q represents sources and sinks of moisture from phase changes and precipitation.

These equations, highly non-linear and coupled, form a closed system that can be solved numerically given appropriate boundary and initial conditions [2].

III-A2 Discretization and Numerical Methods

Since analytical solutions are impossible for these complex PDEs, the continuous equations are discretized onto a computational grid, transforming them into a system of algebraic equations solvable by computers. Common discretization techniques include:

- Finite Difference Methods (FDM): Derivatives are approximated by differences between values at adjacent discrete grid points. This method is straightforward but can suffer from numerical diffusion or dispersion.
- Finite Volume Methods (FVM): Variables are averaged over discrete volumes (cells), and conservation laws are enforced directly across cell interfaces. FVMs are well-suited for ensuring conservation of quantities like mass and energy.
- Spectral Methods: Variables are represented as a sum of basis functions (e.g., spherical harmonics for global models), transforming PDEs into ordinary differential equations (ODEs) for the coefficients. These methods offer high accuracy for smooth solutions but can be computationally expensive for non-smooth features.

Time integration schemes, such as explicit (e.g., Leapfrog), implicit, or semi-implicit methods, are then used to advance the discretized equations forward in time [27]. Explicit schemes are simpler but require small time steps for stability, while implicit/semi-implicit schemes allow larger time steps but are computationally more intensive per step. The choice of resolution, grid type (e.g., latitude-longitude, cubed sphere), and numerical scheme significantly impacts model accuracy, stability, and computational cost.

III-A3 Parameterization of Sub-grid Processes

Processes occurring at scales smaller than the model grid resolution (e.g., clouds, turbulence, convection, radiation, land-surface interactions) cannot be explicitly resolved by the discrete equations. Instead, their collective effects on the resolved scales are represented by parameterization schemes [11]. These schemes use resolved-scale variables (e.g., temperature, humidity at a grid point) to estimate the sub-grid contributions, often relying on empirical relationships, simplified physics, and tunable

parameters. Accurate parameterizations are crucial for realistic forecasts, yet they remain a significant source of model error and uncertainty, as their approximations can be crude and their interactions complex.

III-B Data Assimilation

Data assimilation is the process of combining observations from various sources (satellites, radar, radiosondes, surface stations) with a short-range forecast (the background or first guess) to produce an optimal estimate of the current atmospheric state, known as the analysis [3]. This process is critical because observations are sparse, noisy, and irregularly distributed, while models can drift from reality without continuous correction. The goal is to minimize the difference between the analysis and both the observations and the background, weighted by their respective error covariances.

III-B1 Variational Methods (3D-Var and 4D-Var)

- 3D-Var: Seeks to minimize a cost function $J(\mathbf{x})$ that quantifies the difference between the analysis state $\mathbf{x}$ and the background forecast $\mathbf{x}_b$, and between observations $\mathbf{y}$ and their model equivalent $H(\mathbf{x})$.

$$J(\mathbf{x}) = \frac{1}{2}(\mathbf{x} - \mathbf{x}_b)^T \mathbf{B}^{-1}(\mathbf{x} - \mathbf{x}_b) + \frac{1}{2}(\mathbf{y} - H(\mathbf{x}))^T \mathbf{R}^{-1}(\mathbf{y} - H(\mathbf{x}))$$

where $\mathbf{B}$ is the background error covariance matrix, $\mathbf{R}$ is the observation error covariance matrix, and H is the observation operator that maps model state variables to observation space. The minimization is typically performed using iterative descent algorithms.

- 4D-Var: Extends 3D-Var by assimilating observations over a time window (e.g., 6-12 hours), using the adjoint of the NWP model to propagate information backward in time and optimally adjust the initial state of the window [12]. This method provides a more dynamically consistent analysis by accounting for the temporal evolution of the atmosphere. The cost function becomes:

$$J(\mathbf{x}_0) = \frac{1}{2}(\mathbf{x}_0 - \mathbf{x}_b)^T \mathbf{B}^{-1}(\mathbf{x}_0 - \mathbf{x}_b) + \frac{1}{2} \sum_{i=0}^{N_t} (\mathbf{y}_i - H_i(\mathcal{M}_i(\mathbf{x}_0)))^T \mathbf{R}_i^{-1}(\mathbf{y}_i - H_i(\mathcal{M}_i(\mathbf{x}_0)))$$

where $\mathbf{x}_0$ is the initial state at the start of the window, and $\mathcal{M}_i$ is the NWP model integrated up to time i .

III-B2 Ensemble Kalman Filter (EnKF)

EnKF methods provide an alternative by explicitly representing flow-dependent forecast errors through an ensemble of perturbed forecasts. Observations are used to update each member of the ensemble, with the weights determined by the covariance of the ensemble and observation errors [13]. The analysis update for an ensemble member $\mathbf{x}^{(k)}$ is:

$$\mathbf{x}_a^{(k)} = \mathbf{x}_b^{(k)} + \mathbf{K}(\mathbf{y} - H(\mathbf{x}_b^{(k)}))$$

where $\mathbf{K}$ is the Kalman gain, derived from the background error covariance estimated from the ensemble (e.g., $\mathbf{P}_b =$

$\text{cov}(\mathbf{x}_b^{(k)})$) and the observation error covariance $\mathbf{R}$. EnKF is particularly effective in highly nonlinear systems and for capturing flow-dependent error statistics but can be computationally expensive due to the need to run multiple model integrations.

III-C AI/ML Integration in NWP

The application of AI/ML in NWP spans several areas, from improving specific sub-systems to entirely data-driven forecasting.

III-C1 AI for Parameterization

Deep learning models, particularly Convolutional Neural Networks (CNNs) and Generative Adversarial Networks (GANs), are trained on high-resolution simulation outputs or observations to learn mappings for sub-grid processes [15], [16]. For instance, a neural network can learn to predict the tendencies of temperature, humidity, and momentum due to convection, given the resolved-scale atmospheric state as input. This can replace complex, computationally intensive traditional parameterization schemes.

$$\left. \frac{\partial \mathbf{X}_{resolved}}{\partial t} \right|_{subgrid} = \mathcal{NN}(\mathbf{X}_{resolved}, \mathbf{X}_{auxiliary})$$

where $\mathcal{NN}$ represents the neural network and $\mathbf{X}_{resolved}$ are atmospheric state variables at the resolved scale, and $\mathbf{X}_{auxiliary}$ might include surface properties or radiation fields. The goal is to accelerate computation and improve accuracy by learning from data, rather than relying on hand-tuned approximations.

III-C2 Data-Driven Global Forecasting Models

These models aim to learn the entire atmospheric dynamics from historical data, effectively becoming surrogate models for the atmosphere.

- **Model Architecture:** Modern data-driven models often employ advanced deep learning architectures designed for spatio-temporal data:
 - **U-Nets with Residual Connections:** Used in Pangu-Weather [8], they capture multi-scale features and aid information flow across different atmospheric layers. The U-Net’s encoder-decoder structure is adept at processing image-like grid data.
 - **Graph Neural Networks (GNNs):** Used in GraphCast [9], they operate on irregular grids or graph representations of the Earth, capturing long-range dependencies and overcoming limitations of traditional grid-based CNNs on spherical geometries. Message passing mechanisms enable interaction between distant nodes.
 - **Transformers:** Employed in models like FourCastNet [?], they leverage self-attention mechanisms to learn global correlations and handle long sequences, often combined with CNNs for local feature extraction. This allows the model to weigh

the importance of different regions and variables across the globe.

- **Fourier Neural Operators (FNOs):** Also used in FourCastNet, FNOs learn mappings between infinite-dimensional function spaces, effective for solving PDEs and modeling atmospheric dynamics by operating in the frequency domain, which can be computationally efficient.

- **Data Preparation:** These models are trained on vast datasets of reanalysis data (e.g., ERA5 from ECMWF), which provide a consistent, spatially and temporally complete record of the atmospheric state derived from assimilated observations and model forecasts [28]. ERA5, with its high resolution and extensive variable list, serves as a rich source of atmospheric behavior.
- **Training Objective:** The models are typically trained to minimize a loss function, often Mean Squared Error (MSE), between the model’s predicted state $\hat{\mathbf{x}}_{t+\Delta t}$ and the true state $\mathbf{x}_{t+\Delta t}$ at the next timestep:

$$\mathcal{L} = \frac{1}{N_{samples}} \sum_{i=1}^{N_{samples}} \|\hat{\mathbf{x}}_{t+\Delta t}^{(i)} - \mathbf{x}_{t+\Delta t}^{(i)}\|^2$$

The model learns to predict the state at $t + \Delta t$ given the state at t . This prediction can then be iteratively fed back into the model to generate longer-range forecasts, i.e., $\mathbf{x}_{t+k\Delta t} = F^k(\mathbf{x}_t)$, where F is the learned forecast operator.

III-C3 Physics-Informed Machine Learning (PIML)

PIML combines the data-driven learning capabilities of neural networks with the constraints imposed by physical laws [19], [20]. For atmospheric applications, this might involve adding terms to the loss function that penalize violations of conservation laws (e.g., mass, energy, momentum) or known atmospheric relationships (e.g., geostrophic balance).

$$\mathcal{L}_{total} = \mathcal{L}_{data} + \lambda_{physics} \mathcal{L}_{physics} + \lambda_{boundary} \mathcal{L}_{boundary}$$

where $\mathcal{L}_{data}$ is the standard data-driven loss, $\mathcal{L}_{physics}$ enforces physical consistency using differential operators or conservation laws derived from the primitive equations, and $\mathcal{L}_{boundary}$ ensures boundary condition adherence. The hyperparameter λ controls the weight of the physics constraints. This approach aims to improve the physical realism and generalizability of ML models, especially in data-scarce regions, for extreme events, or when extrapolating beyond training data.

III-C4 AI in Data Assimilation

AI methods are also being explored to enhance data assimilation. This includes using ML to learn the background error covariance $\mathbf{B}$ [29], which is crucial for optimal weighting in variational methods, accelerate the observation operator H , or even develop neural networks that directly map observations to an analysis state. Recurrent neural networks (RNNs) or Transformers could potentially learn

the temporal evolution of errors or assimilate observations sequentially. The goal is to make data assimilation more efficient and robust, particularly for handling high-volume, heterogeneous observational datasets and for improving the initialization of AI-driven forecast models.

The methodology in NWP is constantly evolving, with the integration of AI/ML offering transformative potential to improve efficiency, accuracy, and the ability to model complex atmospheric phenomena across various scales.

IV Results and Discussion

The convergence of traditional Numerical Weather Prediction (NWP) with Artificial Intelligence (AI) and Machine Learning (ML) has led to a rich landscape of results, demonstrating both the significant potential and inherent challenges of these integrated approaches. This section discusses key performance metrics, comparative analyses, and the implications of AI integration.

IV-A Performance Metrics in NWP

Evaluating weather forecasts requires robust metrics that capture accuracy across various atmospheric variables and lead times. These metrics allow for a quantitative comparison between different models and against observations or reanalysis data. Common metrics include:

- **Root Mean Squared Error (RMSE):** A widely used metric that measures the average magnitude of the errors between forecasts (F_i) and observations (O_i). Lower RMSE indicates better accuracy.

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (F_i - O_i)^2}$$

RMSE is particularly sensitive to large errors and is often applied to variables like temperature, wind speed, and geopotential height.

- **Anomaly Correlation Coefficient (ACC):** This metric measures the correlation between the forecast anomaly (deviation from climatology) and the observed anomaly. ACC values range from -1 to 1, with values above 0.6 often considered useful for medium-range forecasting, indicating that the forecast captures the large-scale atmospheric patterns correctly. ACC is frequently used for geopotential height fields.
- **Bias:** Represents the systematic error in the forecast, evaluated as the mean difference between forecast and observation. A persistent positive or negative bias indicates a systematic over- or under-prediction by the model.

$$\text{Bias} = \frac{1}{N} \sum_{i=1}^N (F_i - O_i)$$

- **Skill Scores:** For specific phenomena, particularly precipitation, binary events (e.g., occurrence of heavy rain), or categorized forecasts, skill scores like the

Equitable Threat Score (ETS), Fractions Skill Score (FSS), or Brier Score are employed. These scores evaluate the skill of a forecast relative to a random forecast or climatology, providing a more nuanced measure of predictive capability for rare events.

- **Spatial Verification Metrics:** Beyond point-wise comparisons, spatial metrics like the Structure-Amplitude-Location (SAL) score are used to evaluate forecasts of spatially coherent phenomena (e.g., storm systems), assessing errors in terms of their size, intensity, and position.

These metrics are applied to various atmospheric variables at different levels (e.g., geopotential height at 500 hPa, a proxy for atmospheric circulation; temperature at 850 hPa and 2m; precipitation; and wind speeds at various levels) and across different forecast lead times (e.g., 24-hour, 3-day, 7-day).

IV-B Comparative Analysis: AI vs. Traditional NWP

Recent studies have rigorously compared the performance of state-of-the-art AI-driven weather models against established operational physics-based NWP models, revealing compelling insights into their respective strengths and weaknesses.

IV-B1 Computational Efficiency

One of the most striking advantages of AI models is their computational efficiency during inference. Once trained, models like Pangu-Weather [8], FourCastNet [?], and GraphCast [9] can generate global medium-range forecasts in minutes or even seconds on a single GPU or a small cluster, compared to the hours required by traditional NWP models on supercomputers, which often utilize tens of thousands of CPU cores. For example, a 10-day global forecast from Pangu-Weather can be generated in approximately 1.4 seconds, while a comparable forecast from a leading NWP model might take several hours. This speed allows for more frequent forecast updates, significantly larger and higher-resolution ensemble forecasts, and rapid scenario testing, which is crucial for real-time decision-making and emergency response [18].

IV-B2 Forecast Accuracy

For medium-range global forecasts (typically 3-10 days), data-driven models have shown remarkable accuracy, often matching or exceeding the performance of leading operational physics-based models for certain variables and lead times.

- **Geopotential Height and Temperature:** AI models have demonstrated high ACC for geopotential height at 500 hPa and RMSE for 2m temperature that are competitive with or superior to leading NWP centers for forecasts up to 7-10 days [8], [9]. For instance, Pangu-Weather has shown a 10-day forecast RMSE for 500 hPa geopotential height that is lower than that of the ECMWF IFS for some regions.

- **Precipitation and Extreme Events:** While AI models show strong promise, predicting precipitation, especially extreme events, remains a challenge. Physics-based models often retain an edge due to their detailed microphysical schemes and explicit representation of convective processes. However, AI models are rapidly improving in this domain by leveraging vast datasets and advanced architectures that can capture localized extreme patterns [30]. Some AI models are beginning to show competitive skill for short-term heavy precipitation events [24].
- **Long-term Stability and Climate Drift:** A significant concern with purely data-driven models is their potential for climate drift, where errors accumulate over long integration periods, leading to unphysical states or a departure from observed climate statistics. Physics-based models, constrained by conservation laws and fundamental atmospheric principles, tend to be more stable over extended periods, making them more suitable for climate simulations or very long-range forecasting. Hybrid models aim to mitigate this by incorporating physical constraints, thereby enhancing stability and preventing unphysical drift [19], [20].

IV-C Strengths of AI/ML Approaches

- **Speed and Scalability:** As highlighted, the rapid inference time of AI models enables more frequent forecasts, larger ensembles, and the ability to explore more scenarios rapidly. This is crucial for nowcasting and emergency management.
- **Ability to Learn Complex Non-linearities:** Deep learning excels at identifying intricate, non-linear relationships in high-dimensional meteorological data that might be difficult to explicitly parameterize in traditional models. This can lead to more accurate representations of processes like cloud formation or turbulent mixing.
- **Reduced Development Cycle for Parameterizations:** ML can quickly develop new parameterizations from high-resolution data or simulations, bypassing the extensive expert knowledge, empirical tuning, and linearization assumptions required for traditional schemes [15].
- **Handling Large, Heterogeneous Data:** AI models are adept at processing and learning from diverse observational datasets (satellite imagery, radar returns, ground-based sensor networks) more systematically, potentially extracting more information than traditional assimilation schemes.
- **Potential for Multi-modal Forecasting:** AI can naturally integrate different types of data (e.g., images, time series, gridded data) to produce forecasts for various variables and phenomena simultaneously.

IV-D Weaknesses and Limitations

- **Physical Consistency and Interpretability:** Purely data-driven models may produce physically implausible forecasts, especially for unseen or extreme weather patterns, because they learn correlations rather than underlying physical causality. Their black-box nature also makes it challenging to understand why a specific forecast was made, which hinders scientific insight and user trust [31].
- **Data Dependency and Generalization:** These models require vast amounts of high-quality training data. Their performance can degrade significantly when extrapolating to conditions outside the training distribution, such as unprecedented climate change scenarios, rare extreme events, or new geographical regions for which limited training data exists.
- **Uncertainty Quantification:** While ensemble forecasting is standard in NWP, developing robust and reliable uncertainty quantification methods for AI models is an ongoing challenge. Generating physically meaningful ensemble perturbations and ensuring that the ensemble spread accurately reflects forecast uncertainty are complex tasks.
- **Initial Conditions Sensitivity:** Like traditional models, AI models are sensitive to initial conditions. Current data assimilation techniques are primarily designed for physics-based models, requiring significant adaptation or re-invention for AI systems, especially for models that do not have explicit physical states [29].
- **Resource-Intensive Training:** Although inference is fast, training state-of-the-art AI weather models requires immense computational resources and large datasets, often consuming millions of GPU hours.

IV-E Ensemble Forecasting with AI Models

Ensemble forecasting is crucial for quantifying forecast uncertainty and providing probabilistic predictions, which are often more valuable than single deterministic forecasts. AI models can contribute significantly to this area:

- **High-Resolution Ensembles:** The computational speed of AI models during inference enables running much larger and higher-resolution ensembles (e.g., hundreds or thousands of members), leading to potentially more accurate and reliable probabilistic forecasts [30].
- **Perturbation Generation:** ML can be used to generate intelligent initial condition perturbations or model perturbations that better reflect true forecast uncertainties and growth of errors, rather than relying on simpler statistical methods.
- **Post-processing Ensembles:** AI can refine ensemble outputs, correcting biases, improving calibration of probabilistic forecasts, and extracting high-value information from the ensemble distribution.

The integration of AI into ensemble forecasting holds the promise of dramatically improving the utility of weather predictions by providing more precise and reliable uncertainty estimates.

In summary, AI has emerged as a formidable tool in NWP, capable of delivering highly accurate and efficient forecasts. While still facing challenges related to physical consistency, interpretability, and generalization, the synergistic combination of AI with traditional physics-based models, particularly through hybrid and physics-informed approaches, holds the most promise for advancing the next generation of weather prediction systems. The discussion underscores a shift towards a hybrid paradigm, where the strengths of both methodologies are leveraged to overcome individual limitations.

V Conclusion and Future Work

Numerical Weather Prediction (NWP) has evolved into a highly sophisticated scientific and computational endeavor, providing invaluable forecasts that underpin numerous societal functions. This paper has explored the foundational principles of traditional NWP, including the governing atmospheric equations, numerical discretization methods, and the critical role of data assimilation. We have also delved into the transformative impact of Artificial Intelligence (AI) and Machine Learning (ML) on this field, highlighting how these technologies are being employed to enhance forecast accuracy, computational efficiency, and the representation of complex atmospheric processes.

The advent of purely data-driven weather models, such as Pangu-Weather and GraphCast, marks a significant milestone, demonstrating that AI can achieve forecast skill comparable to, and in some cases surpassing, state-of-the-art physics-based models for medium-range predictions, often with drastically reduced computational costs [8], [9]. Concurrently, AI's role in improving specific NWP components, like sub-grid parameterizations and data assimilation, is proving equally impactful. However, challenges persist, particularly concerning the physical consistency, interpretability, and generalization capabilities of purely data-driven models, especially for extreme events or unprecedented climate conditions.

V-A Future Work

The future of NWP is likely to be characterized by a deeper integration of physical understanding with advanced AI techniques, fostering a hybrid approach that leverages the best of both worlds. Several key directions for future research and development emerge:

V-A1 Advanced Hybrid Models

Developing more sophisticated hybrid models that seamlessly combine physics-based constraints with data-driven learning is paramount. This includes refined Physics-Informed Neural Networks (PINNs) that can enforce conservation laws and thermodynamic consistency

more robustly, potentially through novel architectural designs or advanced loss functions [19], [20]. Research will focus on architectures that allow for dynamic switching between physical and ML components, or that embed physical principles more deeply into the neural network structure, ensuring both accuracy and physical realism. This will help address the limitations of purely data-driven models in terms of physical consistency and stability.

V-A2 Improved Uncertainty Quantification

While AI models can produce forecasts quickly, rigorously quantifying the uncertainty associated with these predictions remains an active area of research. Future work should focus on developing robust AI-native ensemble forecasting techniques, improved perturbation strategies for initial conditions (e.g., using learned patterns of error growth), and methods to accurately estimate prediction intervals and probabilities, moving beyond single deterministic point forecasts [18]. This will be critical for providing actionable information to decision-makers, especially for high-impact weather events where the probability of occurrence is as important as the event itself.

V-A3 Explainable AI (XAI) for NWP

The "black-box" nature of many deep learning models limits their trust and adoption in critical applications. Developing Explainable AI (XAI) techniques tailored for NWP is crucial to understand why an AI model makes a particular forecast, identify potential biases, and gain insights into learned atmospheric relationships [31]. This could involve attribution methods (e.g., saliency maps to highlight influential input regions), sensitivity analysis, or integrating domain expertise into model interpretability frameworks to build confidence in AI-driven predictions. Transparency is key for operational adoption.

V-A4 High-Resolution and Localized Forecasts

Extending the capabilities of AI models to provide high-resolution forecasts at regional and local scales, particularly for nowcasting and urban meteorology, is a significant future direction. This will require incorporating high-fidelity local observations (e.g., from dense urban sensor networks) and developing multi-scale AI architectures that can seamlessly bridge global and local atmospheric dynamics. The challenge lies in efficiently handling vast amounts of localized data and accurately capturing small-scale phenomena like thunderstorms and fog.

V-A5 Data Assimilation for AI Models

Traditional data assimilation schemes are specifically designed for physics-based models. Adapting or re-inventing data assimilation techniques for AI forecasting models is essential. This includes exploring AI-driven observation operators, learning flow-dependent error covariances using ML, and developing methods to assimilate novel, high-volume observational data streams (e.g., from IoT sensors, drones) directly into AI forecast systems [29]. AI could also be used to learn optimal strategies for combining different types of observations or to perform quality control.

V-A6 Long-Term Predictability and Climate Modeling

While current AI models excel in medium-range forecasts, their application to sub-seasonal to seasonal (S2S) prediction and climate modeling is an emerging frontier [26]. Addressing issues like climate drift and ensuring long-term stability and physical consistency will be vital for these extended lead times, potentially through hybrid models or AI-augmented climate simulations. AI could also accelerate climate model components, enabling higher resolution or longer simulations to better understand climate change impacts [24].

In conclusion, the intersection of NWP and AI is a vibrant and rapidly evolving field. By synergistically combining the well-established physical principles of atmospheric science with the unprecedented learning capabilities of AI, we stand on the cusp of a new era in weather forecasting, promising more accurate, efficient, and impactful predictions for the benefit of society. Continued interdisciplinary collaboration between meteorologists, atmospheric physicists, and AI researchers will be crucial to unlock the full potential of these advancements.

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